Dynamic-Routing 3D-Fusion Network for Remote Sensing Image Haze Removal

Hang Sun, Shuanglong Li, Bo Du, *Senior Member, IEEE*, Lefei Zhang, *Senior Member, IEEE*, Dong Ren, and Lyuyang Tong

***Abstract*— Recently, U-shaped neural networks (U-Net) and full resolution convolutional neural networks (F-Net) have been extensively explored for remote sensing image haze removal, achieving excellent performance. However, downsampling in U-Net leads to significant loss of high-frequency information, while F-Net fails to satisfy the large receptive field demand of remote sensing images, resulting in suboptimal dehazing results for both architectures. Moreover, most existing haze removal methods neglect exploring the correlation between spatial and channel information in feature fusion, which is crucial for restoring image texture details and colors. To address these issues, we propose a dynamic-routing 3D-fusion network (DR3DF-Net), comprising a dynamic routing features framework (DRFF) and a 3-D perceptual feature fusion (3D-PFF) module. Specifically, the DRFF utilizes a self-generated constrained feature routing (SCFR) mechanism to learn the most representative features extracted from U-Net, F-Net, and their fused features to enhance clear image reconstruction. Furthermore, the 3D-PFF module enhances interaction between spatial and channel information of multiple features, assigning pixel-level weights for feature fusion, improving dehazed image texture details and colors. Experiments on challenging benchmark datasets demonstrate our DR3DF-Net outperforms several state-of-the-art haze removal methods. The source code is available at https://github.com/lslyttx/DR3DF-Net.**

***Index Terms*— 3-D perceptual feature fusion (3D-PFF), dynamic routing features framework (DRFF), haze removal, remote sensing image.**

I. Introduction

REMOTE sensing images are widely applied in various fields and play a crucial role in geological exploration [1], [2], climate change studies [3], [4], and environmental protection [5], [6], [7]. However, the frequent occurrence of haze in remote sensing images usually results in severe image degradation, such as blurred details and color distortion, restricting their application in advanced computer vision tasks like change detection [8], [9], [10] and object detection [11], [12], [13]. Therefore, the effective removal of haze from remote sensing images to restore clear images has attracted considerable attention from researchers.

Early haze removal methods for natural images and remote sensing images both relied on some hand-crafted priors and atmospheric scattering model (ASM) [14] to restore clear images [15], [16], [17]. However, in complex haze scenarios, the estimated priors are frequently inaccurate, resulting in significantly undesirable dehazed images. In recent years, a substantial number of end-to-end haze removal methods have been proposed, leveraging convolutional neural networks (CNNs) and transformers to learn the mapping between hazy and clear images [18], [19], [20]. The frameworks of these dehazing models primarily include U-shaped neural network (U-Net) and full resolution CNN (F-Net). Most methods employing these two architectures have achieved remarkable improvements in dehazing performance, however, there are still two major issues.

1) *How To Effectively Utilize the Features Extracted From the U-Net and F-Net Architectures for Clear Image Reconstruction:* The U-Net models [21], [22], [23] are widely applied in remote sensing image haze removal. However, the downsampling in the encoding phase can lead to loss of high-frequency information, resulting in poor recovery of small-scale objects under haze. In contrast, the F-Net haze removal models [18], [24], [25] without downsampling can effectively preserve high-frequency feature information, but the limited receptive leads to residual haze. Recently, several dehazing models based on sequential and parallel dual-network architectures have been proposed, demonstrating significant improvements in haze removal. Sequential dual-network models [24], [26], [27] employ the output of the first network as input for the second, resulting in information loss that the second network cannot completely reconstruct. Parallel dual-network models [28], [29], [30] simply concatenate features extracted by U-Net and F-Net for clear image reconstruction, but still exhibit inadequate dehazing performance. These dual-network models fail to effectively utilize features from diverse network architectures for optimal image reconstruction and introduce substantial parameter and information redundancy through stacking existing network structures.
2) *The Feature Fusion in Existing Haze Removal Models Lacks Spatial and Channel Information Interaction Between Multiple Features:* Most existing haze removal

Received 15 December 2024; accepted 2 January 2025. Date of publication 8 January 2025; date of current version 28 January 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62002233. *(Corresponding authors: Dong Ren; Lyuyang Tong.)*

Hang Sun, Shuanglong Li, and Dong Ren are with Hubei Key Laboratory of Intelligent Vision Based Monitoring for Hydroelectric Engineering, College of Computer and Information Technology, China Three Gorges University, Yichang 443002, China (e-mail: sunhang@ctgu.edu.cn; lslyttx@gmail.com; rendong5227@163.com).

Bo Du, Lefei Zhang, and Lyuyang Tong are with the School of Computer Science, Wuhan University, Wuhan 430079, China (e-mail: dubo@whu.edu.cn; zhanglefei@whu.edu.cn; lyuyangtong@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3526993

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.

networks mainly utilize element-wise addition [21], [25], [31] and channel concatenation [32], [33], [34] for feature fusion. In recent years, various feature fusion methods leveraging learnable parameters [19], [35] and attention mechanisms [18], [36], [37] have been proposed to enhance haze removal model performance. These methods allocate weights to the spatial, channel, and overall aspects of features, enabling effective constraints on the fusion process and facilitating more accurate feature fusion. However, these methods suffer from coarse fusion granularity and lack of fine-grained interactions between spatial and channel information, leading to insufficient representational capacity of fused features. This reduces the restoration competencies of the dehazing model in texture details and colors. As a result, delving into the intrinsic spatial and channel correlations among multiple features during feature fusion, achieving fine-grained weight allocation, and enhancing the representation of the fused features to generate outstanding restoration of texture details and colors remains an unresolved issue.

To address the above mentioned issues, in this article, we propose a dynamic-routing 3D-fusion network (DR3DF-Net) for remote sensing image haze removal. Specifically, we propose the dynamic routing features framework (DRFF), which dynamically routes features extracted by a U-Net with large receptive fields, an F-Net with high-frequency details, and their fused features for clear image reconstruction by predicting reconstruction difficulty score of features through a self-generated constrained feature routing (SCFR) mechanism, thus enhancing dehazing performance, particularly for small-scale target regions. Furthermore, the proposed 3-D perceptual feature fusion (3D-PFF) 3-D perceptual feature fusion module involves coarse fusion and fine-grained fusion. Coarse fusion initially aggregates spatial and channel information to learn the global context of each input feature. Subsequently, fine fusion achieves 3D interaction across spatial and channel dimensions through self-replication, generating pixel-level granularity weight matrices allocated to corresponding input features for the second fusion. This two-stage fusion enhances the fused feature representation, improving the restoration of texture details and colors.

The main contributions of this article are summarized as follows.

1) With the well-designed DRFF, self-generated label constrained feature routing (SCFR) mechanism can be introduced to dynamically route representative features extracted from U-Net and F-Net and their fused features for significant improvement in haze removal, especially for small-scale objects under haze in remote sensing images.
2) A 3D-PFF module is proposed to explore how to incorporate the spatial and channel dimensions to effectively preserve significant global context for coarse fusion, and further generate pixel-level granularity weight matrices through self-replicating 3-D interaction for final fusion. Hence, the 3D-PFF module is capable of enhancing the fused feature representation, improving the restoration of texture details and colors.
3) We propose a novel method called DR3DF-Net that seamlessly integrates well-designed DRFF and 3D-PFF. With more effective dynamic feature selection and fused feature representation, experiments on challenging datasets including SateHaze1k [38], RICE [39], and RSID [40] demonstrate our remote sensing haze removal method achieves superior performance compared to several state-of-the-art methods.

The rest of this article is organized as follows. Section II presents the related work. Section III introduces the proposed method, including the DRFF and the 3D-PFF module. Section IV evaluates the proposed method through a series of experiments and Section V discusses the limitation and future work. Finally, Section VI concludes the whole paper.

II. Related Work

This section provides a concise overview of recent advancements in image haze removal methods. In Section II-A, we introduce prior-based haze removal methods, followed by a discussion of supervised learning haze removal methods in Section II-B. Finally, Section II-C reviews unsupervised learning image haze removal methods.

A. Prior-Based Haze Removal Methods

Prior-based haze removal methods typically rely on the ASM [14] and various handcrafted priors, such as the dark channel prior (DCP) [15], nonlocal prior [17], haze-line prior [16], and color attenuation prior [41], to restore clear images. Specifically, He et al. [15] proposed the DCP, which assumes at least one channel for each pixel has a value close to zero. In the field of remote sensing, Pan et al. [42] developed the deformable haze imaging model (DHIM) to estimate atmospheric light and transmission by combining a translation term with the DCP. Li et al. [43] proposed a two-stage haze removal method based on homomorphic filtering (HF) and a spherical model to improve the DCP. Although prior-based methods have achieved some dehazing results, the estimated priors are often inaccurate when dealing with complex natural scenes, leading to suboptimal dehazing performance.

B. Supervised Learning Haze Removal Methods

Early supervised learning haze removal methods [44], [45], [46] utilize CNNs to estimate the atmospheric light and transmission map to reconstruct clear images. For instance, Li et al. [47] designed an AOD-Net to generate restored images by reformulating the ASM [15]. Zhang and Patel [48] embedded the ASM into the network, enabling the network to jointly estimate the transmission map, atmospheric light, and dehazed image.

In recent years, end-to-end haze removal networks based on U-Net [21], [22], [49] or F-Net [18], [24], [25] have become mainstream, most researchers improve haze removal performance by designing different structures and modules within the network. For instance, Chen et al. [19] developed a content-guided attention to emphasize more useful

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.

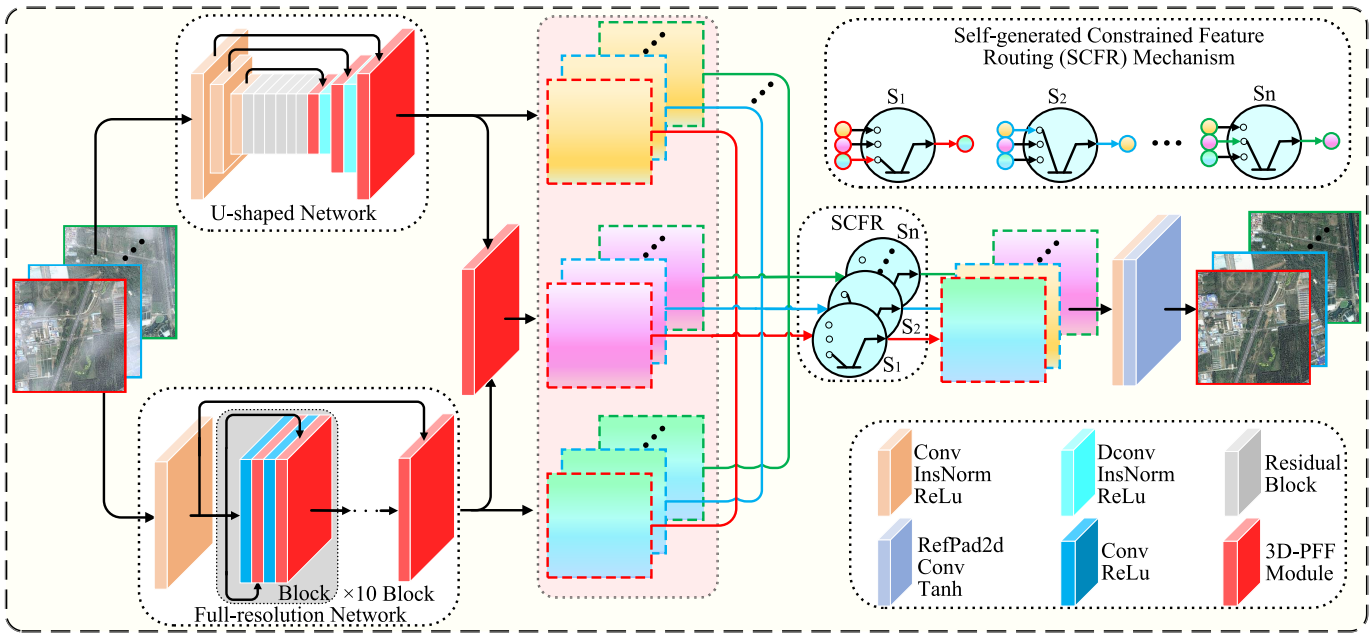


Fig. 1. Overall architecture of our proposed DR3DF-Net. The DRFF can dynamically route features extracted via U-Net and F-Net which contain multiple representative feature information for more optimal clear image reconstruction by incorporating the SCFR mechanism. The 3D-PFF module can enhance the fused feature representation by perceiving global context and generating pixel-level weight matrices during coarse-to-fine fusion, leading to superior texture detail and color restoration. Then, by integrating the DRFF and 3D-PFF module into a dynamic routing 3-D fusion network, our DR3DF-Net outperforms existing haze removal methods.

information encoded in features. Qin et al. [18] designed a feature fusion attention network without downsampling to focus on more important information. Lihe et al. [50] build a physic-aware network to achieve dehazing results with physical interpretability. With the growing potential of Vision Transformers (ViT) in various visual tasks, an increasing number of Transformer-based dehazing models are being developed. For instance, Song et al. [51] proposed DehazeFormer, incorporating ViT as the fundamental block of U-Net. Wang et al. [52] developed a self-prompt dehazing Transformer, exploring depth consistency of hazy images and their corresponding clear images as a prompt. Zhang et al. [53] proposed the proxy and cross-stripes integration transformer (PCSformer) to directly model long-range dependencies and generate more consistent and realistic scene details. However, due to loss of high-frequency information during U-Net encoding and insufficient receptive field of F-Net for remote sensing images, the dehazing performance of both models is unsatisfactory. Therefore, some approaches [24], [27], [28], [29] combine U-Net and F-Net in a serial or parallel manner to improve dehazing results. Jiang et al. [24] proposed a hybrid network consisting of two serial sub-networks, with one focusing on dehazing and the other on refining image details. Yu et al. [28] proposed a parallel dual-branch dehazing framework to integrate the detailed information preserved by F-Net into U-Net.

C. Unsupervised Learning Haze Removal Methods

Motivated by the success of CycleGAN [54] in unpaired image-to-image translation, several researchers attempt to develop unsupervised frameworks for image dehazing [55], [56], [57]. Engin et al. [55] introduced an enhanced CycleGAN framework incorporating perceptual consistency loss to strengthen network constraints. Yang et al. [56] introduced a self-argumentation strategy by modeling the scattering coefficient and depth map of hazy scenes. Zheng et al. [57] presented a Transformer-Guide CycleGAN framework generative adversarial networks, which can extend the applicability of the transformer architecture by incorporating total variation loss (TV Loss). Recently, Li et al. [58] proposed a self-supervised haze removal framework by employing three joint subnetworks to estimate dehazed images, transmission map, and atmospheric light. Sun et al. [59] designed an unsupervised multibranch network to address generator learning confusion in CycleGAN-based methods by disentangling transmission map, atmospheric light, and dehazed images from encoding features.

III. Proposed Method

In this section, we first introduce the overview in Section III-A, then describe the proposed DRFF in Section III-B. After that, we present the 3D-PFF module in Section III-C, and finally explain the loss function used in DR3DF-Net in Section III-D.

A. Overview

The overall architecture of our DR3DF-Net is shown in Fig. 1. For a set of input hazy images, each image is initially processed through a parallel dual-branch network to extract three distinct types of features: the output features from U-Net, the output features from F-Net, and a third type of feature obtained by fusing the first two. Since these three types of features possess different representative information, the model should learn to select the most suitable features to achieve a more ideal reconstruction of clear images. Therefore, we propose the DRFF, which can dynamically select the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.

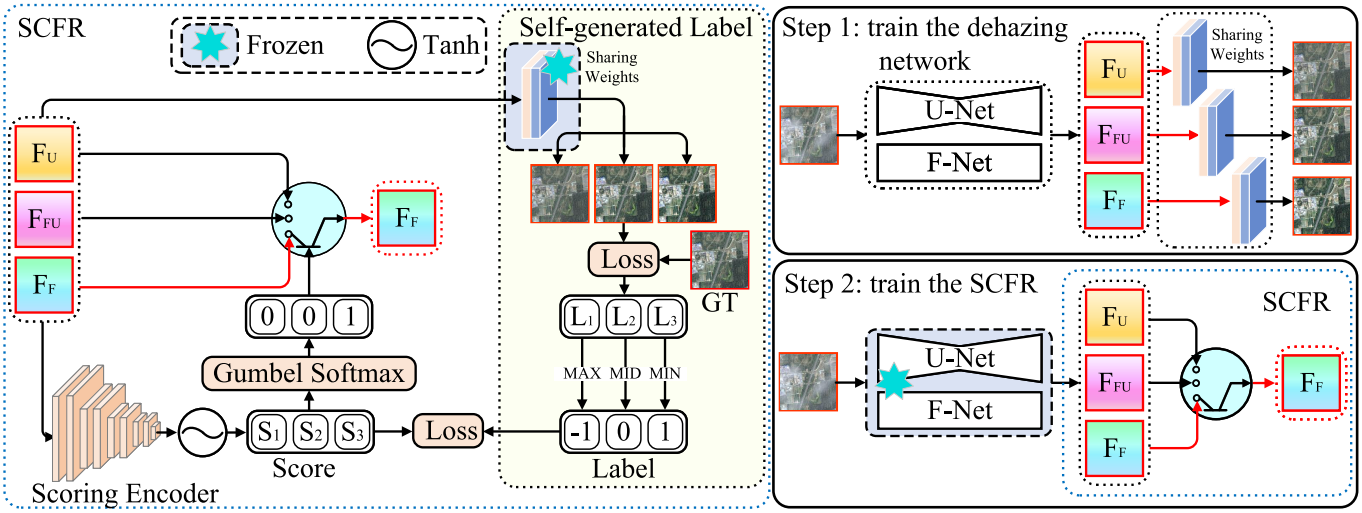


Fig. 2. Overall structure of SCFR mechanism. By constraining the score encoder with self-generated difficulty labels, our SCFR can predict reconstruction difficulty scores for features, dynamically selecting the most appropriate features extracted from U-Net, F-Net, and their fused for each input to recover a clear image during inference. Steps 1 and 2 illustrate the optimization strategy of the proposed DRFF.

most representative features through a SCFR mechanism and utilize trailing convolutional blocks to restore clear images. Additionally, we have designed a 3D-PFF module to facilitate information fusion in the corresponding encoding and decoding layers of U-Net, the fusion of residual features in F-Net, as well as the integration of output features from both models. Integrating the 3D-PFF module into our DR3DF-Net allows the model to more accurately preserve useful features during feature fusion, enhancing its ability to perceive texture details and color, and significantly improving dehazing performance.

B. Dynamic Feature Routing Framework

Recent studies [24], [27], [28] have combined U-Net and F-Net in a serial or parallel manner for more effective haze removal. However, serial methods face challenges in reconstructing information lost in the first stage, while parallel methods simply concatenate features extracted from both networks, leading to suboptimal dehazing performance. How to effectively utilize features extracted from U-Net and F-Net for high-quality clear image reconstruction is still challenging. Hence, in this article, a DRFF is proposed to automatically select the most representative features for each hazy image to reconstruct a clear image by incorporating a SCFR mechanism.

As shown in Fig. 1, hazy images are input into a dual-branch network to extract three types of features. Then, SCFR will be fed with these features to select the most appropriate ones to reconstruct a clear image. As the core of DRFF, the overall structure of our SCFR Mechanism is shown in Fig. 2. Specifically, the SCFR mainly consists of a self-generated label and a scoring encoder. The self-generated label generates corresponding difficulty labels by calculating the reconstruction loss for each feature, constraining the scoring encoder to learn to predict reconstruction difficulty scores. During inference, the Gumbel-Softmax function is utilized to select the most suitable features based on these scores. The optimization strategy of our DRFF is divided into two steps.

Step 1 Train the Dehazing Network: We first train the dehazing network to enable the dual-branch network to efficiently extract features

$$F_{\mathrm{U}}, F_{\mathrm{FU}}, F_{\mathrm{F}} = \mathcal{N}_{\mathrm{dehaze}}^{\mathrm{w/o\ SCFR}}(x, y|\Theta_1) \tag{1}$$

where $\mathcal{N}_{\mathrm{dehaze}}^{\mathrm{w/o\ SCFR}}$ denotes our trained dehazing network without SCFR. x and y denote the hazy image and the ground truth (GT), respectively. Θ denotes the learnable parameters of the dehazing network. F_{U}, F_{FU}, and $F_{\mathrm{F}} \in R^{64\times H\times W}$ represent the features extracted by U-Net, the features jointly extracted by U-Net and F-Net, and the features extracted by F-Net, respectively.

Step 2 Train the SCFR:

After the above training, the dual-branch network is able to extract three representative features from hazy images. We freeze the parameters of the dehazing network to focus on training the SCFR. First, we extract features to be selected

$$F_{\mathrm{U}}, F_{\mathrm{FU}}, F_{\mathrm{F}} = \mathcal{N}_{\mathrm{dehaze}}\left(x|\Theta_1^*\right) \tag{2}$$

where $\mathcal{N}_{\mathrm{dehaze}}(x|\Theta_1^*)$ represents the hazy image input to the dehazing network with frozen parameters for feature extraction. Then, these features will be input into the scoring encoder ($\mathcal{SE}$) to predict a reconstruction difficulty score $S \in (-1, 1)$ for the image

$$S_U, S_{FU}, S_F = \mathrm{Tanh}(\mathcal{SE}(F_{\mathrm{U}}), \mathcal{SE}(F_{\mathrm{FU}}), \mathcal{SE}(F_{\mathrm{F}})) \tag{3}$$

where Tanh denotes the Tanh activation function, mapping the input variable to the range $(-1,1)$. S_U, S_{FU}, and S_F represent predicted reconstruction difficulty scores.

For F_{U}, F_{FU}, and F_{F}, we hope that the scores predicted by $\mathcal{SE}$ for the most suitable features to reconstruct a clear image are close to "1," the suboptimal features are close to "0," and the least suitable features are close to "−1."

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.

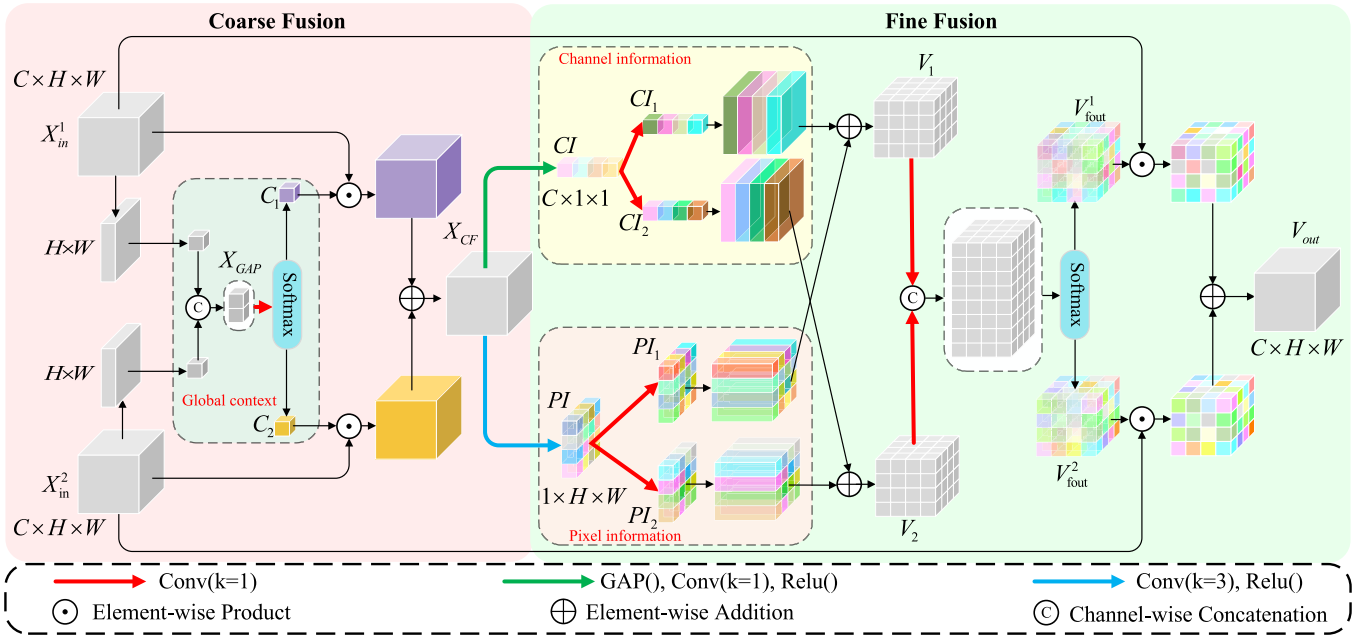


Fig. 3. Overall structure of the 3D-PFF module, which mainly contains coarse fusion and fine fusion. Coarse fusion can effectively preserve global context information by learning the importance of features. Then, with the self-replicating 3-D interaction across spatial and channel dimensions, fine fusion can generate pixel-level granularity weight matrices, thus enhancing feature representation and further improving the restoration of texture details and colors.

In other words, features with higher scores are more likely to reconstruct an ideal clear image. However, the absence of paired labels makes it challenging to train the score encoder to learn this score mapping relationship. Therefore, we designed a self-generating label method, where clear images are introduced to calculate the reconstruction loss for each feature. Based on the loss, labels of "1," "0," and "−1" are generated in ascending order, thereby constraining the $\mathcal{SE}$ and Tanh to learn the feature reconstruction difficulty score mapping

$$L_{\mathrm{U}}, L_{\mathrm{FU}}, L_{\mathrm{F}} = T_{S\to L}(l_{\mathrm{U}}, l_{\mathrm{FU}}, l_{\mathrm{F}}) \tag{4}$$

where L_{U}, L_{FU}, and L_{F} represent the generated labels, l_{U}, l_{FU}, and l_{F} denote the losses for reconstructing clear images using three features, and $T_{S\to L}$ represents the transformation from loss values to labels. The highest loss among the three generates a "−1" label, the smallest generates a "1" label, and the middle generates a "0" label. Subsequently, the generated labels are used to train the $\mathcal{SE}$ and Tanh to learn the correct mapping of reconstruction difficulty scores

$$S_{\mathrm{U}}, S_{\mathrm{FU}}, S_{\mathrm{F}} = \mathcal{N}_{\mathrm{SCFR}}^{\text{w/o dehaze}}(F, L|\Theta_2) \tag{5}$$

where $\mathcal{N}_{\mathrm{SCFR}}^{\text{w/o dehaze}}$ denotes our trained SCFR without dehazing network. F represents the three features F_1, F_2, and F_3, L denotes the corresponding labels, and Θ_2 represents the parameters of SCFR. S_{U}, S_{FU}, and S_{F} denote the difficulty scores predicted by $\mathcal{SE}$. We use L1 loss to constrain the predicted scores and labels. Finally, S_{U}, S_{FU}, and S_{F} are passed to the Gumbel-Softmax function to complete the feature selection, which maps the scores into a binary vector composed of "0" and "1" (B_{U}, B_{FU}, B_{F}), where only the element with the highest score returns as "1," and the rest are "0." This binary vector is then multiplied by the corresponding features, thereby achieving feature routing

$$F = (F_{\mathrm{U}}, F_{\mathrm{FU}}, F_{\mathrm{F}}) \cdot (B_{\mathrm{U}}, B_{\mathrm{FU}}, B_{\mathrm{F}}) \tag{6}$$

where F represents the selected feature. By incorporating the self-generated label constrained feature routing mechanism into the DRFF, it is able to dynamically route features extracted by U-Net, F-Net, and their fused features to reconstruct superior haze-free images, thereby enhancing the recovery of small-scale objects under haze occlusion.

C. 3D-PFF Module

Many haze removal methods [19], [35], [40] have utilized learnable parameters and attention mechanisms for feature fusion, enhancing network performance. However, these methods fail to exploit the intrinsic correlation between spatial and channel information during fusion, resulting in insufficient feature representation. Therefore, in this article, we propose a 3D-PFF module to effectively correlate spatial and channel information during feature fusion in a coarse-to-fine manner for learning more powerful feature representations. The overall structure of the proposed 3D-PFF module is shown in Fig. 3, which includes two stages: coarse fusion and fine fusion.

1) Coarse Fusion: Given features X_{in}^1 and $X_{\mathrm{in}}^2 \in R^{C\times H\times W}$ as input. By learning the importance of features, coarse fusion can effectively preserve crucial global context information. The process is illustrated as follows:

$$X_{\mathrm{GAP}} = \mathrm{Cat}\left(\mathrm{P}_{\mathrm{s,c}}\left(X_{\mathrm{in}}^1\right), \mathrm{P}_{\mathrm{s,c}}\left(X_{\mathrm{in}}^2\right)\right) \tag{7}$$

where $\mathrm{P}_{s,c}$ represents global average pooling, which aggregates spatial and channel information to obtain representative descriptors X_{GAP}. Cat denotes channel concatenation. Subsequently, a 1×1 convolution and the Softmax function are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.

applied to transform X_{GAP} into the importance weights of the two features

$$C_1, C_2 = \mathrm{Softmax}(\mathrm{Conv}(X_{\mathrm{GAP}})) \tag{8}$$

where C_1, C_2 indicates the importance weights. Finally, with the feature weights C_1 and C_2, the coarse fused features are obtained by

$$X_{\mathrm{CF}} = X_{\mathrm{in}}^1 * C_1 + X_{\mathrm{in}}^2 * C_2 \tag{9}$$

where $*$ denotes element-wise product. With coarse fusion, we can learn effective importance weights of features for preserving crucial global context information.

2) Fine Fusion: To fully explore the correlation between spatial and channel information for enhanced representation capabilities, we further adaptively calibrate the retention of important pixel-level feature information by explicitly modeling the interdependencies between spatial and channel dimensions, i.e., a fine fusion stage is proposed to generate pixel-level 3-D weight matrices through self-replicating 3-D interactions of spatial description map and channel description vector. We take the feature X_{CF} from coarse fusion as input for fine to selectively emphasize critical pixel-level feature information and flexibly handle haze-related features.

First, we extract the spatial description maps PI and channel description vectors CI from X_{CF}. During extraction of pixel description maps, we employed a convolution layer and an activation function to effectively extract location-related information features. By utilizing global average pooling to compress spatial information, and subsequently employing a convolutional layer and an activation function, we can efficiently extract channel description vectors. Then, a 1×1 convolution is utilized to self-replicate CI and PI, enabling more accurate learning of the spatial description map and channel description vector for each original input feature

$$\begin{aligned} \mathrm{PI}_1, \mathrm{PI}_2 &= \mathrm{Conv}(\mathrm{PI}), \mathrm{Conv}(\mathrm{PI}) \\ \mathrm{CI}_1, \mathrm{CI}_2 &= \mathrm{Conv}(\mathrm{CI}), \mathrm{Conv}(\mathrm{CI}) \end{aligned} \tag{10}$$

where PI_1, PI_2 and CI_1, CI_2 represent the spatial description map and the channel description vector for each original input. On this basis, we extend the dimensions of the spatial description map and the channel description vector by using broadcasting mechanism, and fuse the two types of information through element-wise addition and a 1×1 convolution. The resulting 3-D description matrix can establish the dependency between spatial and channel information for each pixel

$$\begin{aligned} V_1 &= \mathrm{Conv}\left(\mathrm{F}_{\mathrm{brd}}\left(\mathrm{CI}_1\right) + \mathrm{F}_{\mathrm{brd}}\left(\mathrm{PI}_1\right)\right) \\ V_2 &= \mathrm{Conv}\left(\mathrm{F}_{\mathrm{brd}}\left(\mathrm{CI}_2\right) + \mathrm{F}_{\mathrm{brd}}\left(\mathrm{PI}_2\right)\right) \end{aligned} \tag{11}$$

where $F_{\mathrm{brd}}()$ represents the broadcast mechanism. Subsequently, a Softmax function is utilized to map V_1 and V_2 into pixel-level weight matrices for each feature and the final process can be expressed by the following equation:

$$V_{\mathrm{fout}}^1, V_{\mathrm{fout}}^2 = \mathrm{Softmax}(\mathrm{Cat}(V_1, V_2)) \tag{12}$$

$$V_{\mathrm{out}} = X_{\mathrm{in}}^1 * V_{\mathrm{fout}}^1 + X_{\mathrm{in}}^2 * V_{\mathrm{fout}}^2 \tag{13}$$

where V_{out} represents the final fused feature. V_{fout}^1 and V_{fout}^2 are generated pixel-wise weight matrices. Each weight element simultaneously considers both the spatial location and channel importance of the pixel, as well as the global context, which makes the network focus more on preserving useful features, thereby enhancing feature representation and significantly improving texture detail and color restoration in dehazed images.

D. Loss Function

During the training phase, we utilize smooth L1 loss, adversarial loss, perceptual loss, and multiscale structural similarity (MS-SSIM) loss to optimize our dehazing network.

1) Smooth L1 Loss: Smooth L1 loss is absolute losses and is sensitive to outliers. Our DR3DF-Net uses Smooth L1 loss as the constraint for dehazed image reconstruction

$$L_{\mathrm{smoothL1}} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{SmoothL1}\left(z_i - G\left(x_i\right)\right) \tag{14}$$

$$\mathrm{SmoothL1}(i) = \begin{cases} 0.5i^2, & if \quad |i|<1 \\ |i| - 0.5, & \mathrm{else} \end{cases} \tag{15}$$

where G denotes the generator of the DR3DF-Net and N represents the number of pixels in the image. x_i and z_i denote the values of pixel i in the hazy image and the GT, respectively.

2) Adversarial Loss: We utilize the adversarial loss of GANs to constrain the network in generating clearer images. The specific definition of adversarial loss is as follows:

$$L_{\mathrm{adv}} = \sum_{n=1}^{N} \log(1 - D(G(X))) \tag{16}$$

where G denotes the DR3DF-Net and D represents the discriminator.

3) Perceptual Loss: Perceptual Loss [60] is based on the perceptual quality of images, focusing more on advanced features. It effectively calculates the L2 norm at different layers of the VGG16 [61] pretrained model for GT and dehazed images from the network. The definition of perceptual loss in this study is as follows:

$$L_{\mathrm{perc}} = \frac{1}{N}\sum_{i} \frac{1}{C_i H_i W_i} \|\eta_i(Y) - \eta_i(G(X))\|_2^2 \tag{17}$$

where η_i represents the size of the feature map obtained from layer i of the pretrained VGG16 model. N represents the number of feature layers used in the perceptual loss. X and Y represent the hazy image and the GT, respectively.

4) MS-SSIM Loss: This study incorporates the MS-SSIM to promote closer structural similarity (SSIM) between dehazed and clear images. The MS-SSIM is illustrated as follows:

$$L_{\mathrm{ms\text{-}ssim}} = 1 - \prod_{m=1}^{M} \left(\frac{2\varphi_i\varphi_j + C_1}{\varphi_i^2 + \varphi_j^2 + C_1}\right)^{\alpha_m} \left(\frac{2\eta_{ij} + C_2}{\eta_i^2 + \eta_j^2 + C_2}\right)^{\beta_m} \tag{18}$$

where φ_i and φ_j denote the mean of the dehazed image and its GT, respectively. η_{ij} denotes the covariance of the dehazed image and its GT, η_i and η_j denote the standard deviation of the dehazed image and its GT. C is a constant.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE COMPARISONS OF VARIOUS HAZE REMOVAL METHODS ON THE SATEHAZE1K DATASET. THE RED, GREEN, AND BLUE FONTS REPRESENT THE FIRST, SECOND, AND THIRD RANKINGS IN TERMS OF PSNR OR SSIM

Methods	Venue&Year	Thin		Moderate		Thick		Average		Params.↓
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	
DCP [15]	TPAMI, 2010	13.45	0.701	9.780	0.592	10.89	0.572	11.37	0.621	17.37M
FFA [18]	AAAI, 2020	23.75	0.903	26.50	0.941	22.03	0.840	24.09	0.894	4.686M
FCTF [27]	GRSL, 2021	20.03	0.880	24.19	0.929	19.18	0.832	21.16	0.880	0.16M
TBN [28]	CVPRW, 2021	24.52	0.897	27.42	0.944	21.56	0.837	24.50	0.892	50.35M
Restormer [62]	CVPR, 2022	23.08	0.912	24.73	0.903	19.58	0.792	22.46	0.869	26.10M
SGID [63]	TIP, 2022	23.34	0.907	23.95	0.935	19.63	0.827	22.30	0.889	13.87M
DCI-Net [22]	TGRS, 2022	24.47	0.882	25.05	0.910	22.73	0.806	24.08	0.866	26.51M
DehazeFormer [51]	TIP, 2023	24.13	0.917	26.53	0.941	24.68	0.875	25.04	0.908	2.514M
MFINEA [20]	NN, 2023	26.16	0.912	27.61	0.942	23.91	0.841	25.89	0.898	9.49M
Trinity-Net [40]	TGRS, 2023	22.66	0.815	25.02	0.929	22.05	0.825	23.24	0.856	20.14M
PSMB-Net [49]	TGRS, 2023	26.75	0.928	27.48	0.946	25.15	0.889	26.46	0.921	12.49M
PCSformer [53]	TGRS, 2024	26.47	0.923	28.06	0.943	24.50	0.871	26.34	0.912	3.73M
DEA-Net [19]	TIP, 2024	26.72	0.931	27.68	0.952	24.74	0.873	26.38	0.918	3.65M
Ours	-	**27.70**	**0.935**	**29.04**	**0.956**	**25.72**	**0.902**	**27.48**	**0.931**	9.08M

The total loss function of the network is defined as follows:

$$L = \lambda_1 L_{\text{smoothL1}} + \lambda_2 L_{\text{adv}} + \lambda_3 L_{\text{pere}} + \lambda_4 L_{\text{ms-ssim}} \tag{19}$$

where λ_1, λ_2, λ_3, and λ_4 are hyperparameters that balance between different losses.

IV. EXPERIMENTAL RESULTS

We first introduce the experimental settings in Section IV-A. Then, we present the quantitative and qualitative evaluation results of DR3DF-Net with excellent haze removal methods (DCP [15], FFA [18], FCTF [27], TBN [28], Restormer [62], SGID [63], DCI-Net [22], MFINEA [20], DehazeFormer [51], Trinity-Net [40], PSMB-Net [49], PCSformer [53], and DEA-Net [19]) in Sections IV-B and IV-C, respectively. Finally, we conduct ablation studies in Section IV-D to verify the effectiveness of each module.

A. Experimental Settings

Datasets: In our implementation, we evaluate the proposed DR3DF-Net using three publicly available remote sensing datasets: SateHaze1k [38], RSID [40], and RICE [39]. The SateHaze1k dataset consists of three subsets: thin, moderate, and thick, each with 400 synthesized remote sensing image pairs. There are 320 pairs in the training set, 35 in the validation set, and 45 in the test set per subset. Images have a resolution of 512 × 512 pixels. The RSID dataset contains 1000 synthesized image pairs, with 256 × 256 resolution. We randomly select 900 pairs for training and reserve the rest for testing. The RICE dataset has two subsets: RICE1 and RICE2, both real-world remote sensing datasets. The RICE1 dataset has 500 image pairs at 512 × 512 resolution collected from Google Earth. The RICE2 dataset has 450 pairs at the same resolution, collected from Landsat8 OLI/TIRS. We randomly divided the entire RICE1 dataset into a training set and a test set at a ratio of 9:1 and the RICE2 dataset is similarly divided.

Training Details: We employed the Adam optimizer with momentum decay indices set to $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The learning rate and batch size were set to 0.0001 and 4, respectively. The initial learning rate was 0.001, which was dynamically adjusted using MultiStepLR. Throughout training, the parameters of the loss functions were set to $\lambda_1 = 1$, $\lambda_2 = 0.0005$, $\lambda_3 = 0.01$, and $\lambda_4 = 0.5$. All experiments in this study were conducted on an NVIDIA RTX 3090 GPU, with code implementation via PyTorch.

Evaluation Metrics: We adopt the peak signal noise ratio (PSNR) [64] and the SSIM to quantitatively evaluate the dehazing results of all algorithms. These two metrics are commonly used to measure image quality in computer vision. A higher PSNR and SSIM value indicates that the dehazed images are closer to the corresponding GT.

B. Quantitative Evaluations

We quantitatively evaluate the proposed DR3DF-Net with other excellent haze removal methods on three public remote sensing datasets. Tables I and II present the PSNR and SSIM metrics for all methods. The red, green, and blue fonts indicate the first, second, and third rankings in terms of PSNR or SSIM values. Additionally, Table I compares the number of parameters of each method.

Results and analysis on StateHaze1K dataset. Table I presents quantitative comparisons of various algorithms on the SateHaze1K dataset. Experimental results demonstrate that the proposed DR3DF-Net achieves the best performance across all three subsets: thin haze, moderate haze, and thick haze. Specifically, DR3DF-Net reaches a PSNR of 27.70 dB and SSIM of 0.935 under thin haze, 29.04 dB and 0.956 under moderate haze, and 25.72 dB and 0.902 under thick haze. Compared to state-of-the-art dehazing algorithms such as DCI-Net, PCSformer, and DEA-Net, our method achieves an average PSNR improvement of 3.4, 1.14, and 1.1 dB across the three subsets, with corresponding SSIM gains of 0.065, 0.019, and 0.013. While PCSformer and DEA-Net achieve comparable results on certain subsets, they fail to consistently improve across all subsets. This is mainly attributed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.

TABLE II
QUANTITATIVE COMPARISONS OF VARIOUS HAZE REMOVAL METHODS ON THE RSID AND RICE DATASETS. THE RED, GREEN, AND BLUE FONTS REPRESENT THE FIRST, SECOND, AND THIRD RANKINGS IN TERMS OF PSNR OR SSIM

Methods	Venue&Year	RSID		RICE1		RICE2		Average	
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
DCP [15]	TPAMI, 2010	16.99	0.804	12.39	0.628	12.47	0.535	13.95	0.655
FFA [18]	AAAI, 2020	25.70	0.949	35.01	0.951	34.82	0.894	31.84	0.931
FCTF [27]	GRSL, 2021	21.38	0.890	32.06	0.941	31.44	0.859	28.29	0.896
TBN [28]	CVPRW, 2021	27.27	0.947	35.80	0.959	34.67	0.876	32.58	0.927
Restormer [62]	CVPR, 2022	26.91	0.942	27.92	0.931	24.61	0.801	26.48	0.894
SGID [63]	TIP, 2022	25.25	0.945	29.73	0.947	28.23	0.833	27.73	0.908
DCI-Net [22]	TGRS, 2022	24.86	0.944	35.03	0.960	32.14	0.886	30.67	0.930
DehazeFormer [51]	TIP, 2023	27.78	0.942	35.49	0.957	33.17	0.890	32.17	0.929
MFINEA [20]	NN, 2023	25.50	0.950	32.08	0.956	32.56	0.886	30.04	0.930
Trinity-Net [40]	TGRS, 2023	27.13	0.941	27.30	0.913	24.71	0.794	26.38	0.882
PSMB-Net [49]	TGRS, 2023	26.60	0.948	34.68	0.963	33.96	0.893	31.74	0.934
PCSformer [53]	TGRS, 2024	28.69	0.957	36.09	0.954	34.35	0.885	33.04	0.932
DEA-Net [19]	TIP, 2024	27.58	0.959	35.44	0.962	34.61	0.896	32.53	0.939
Ours	-	**29.74**	**0.963**	**37.43**	**0.965**	**35.76**	**0.898**	**34.31**	**0.942**

to our proposed dynamic feature routing framework, which allows the network to adaptively select optimal reconstruction features for each sample, demonstrating strong generalization capability across varying haze distributions. Additionally, our 3D-PFF module can effectively preserve critical global context, spatial, and channel information during feature fusion, enhancing representational capacity and significantly boosting dehazing performance. Table I also compares model parameters. Our method has fewer parameters than recent remote sensing dehazing algorithms such as DCI-Net, PSMB-Net, and Trinity-Net. Although it has slightly more parameters than PCSformer and DEA-Net, our approach achieves substantially better PSNR and SSIM performance than both PCSformer and DEA-Net.

Results and analysis on RSID and RICE datasets. As shown in Table II, the proposed DR3DF-Net outperforms all other algorithms across all metrics on both the RSID and RICE datasets. On the RICE1 and RICE2 real-world remote sensing datasets, our DR3DF-Net achieves PSNR values of 37.43 and 35.76 dB, respectively, with SSIM values of 0.965 and 0.898, demonstrating strong generalization capability in real-world scenarios. Additionally, on the RSID dataset, the PSNR is 29.74 dB and the SSIM is 0.963, further validating the effectiveness of the model. Methods such as FFA, DehazeFormer, PCSformer, DEA-Net, and TBN, which are based on F-Net and U-Net, perform well on some datasets but struggle to achieve consistent results across all datasets. Notably, TBN simply concatenates features extracted by two types of networks, making it difficult to achieve stable performance improvements. In contrast, our DR3DF-Net is capable of automatically selecting the best features extracted by two models for image reconstruction, thereby enhancing dehazing performance. Furthermore, by considering spatial and channel correlations during feature fusion, enables pixel-level weight allocation, enhancing feature representation and improving texture details and color restoration.

C. Qualitative Analysis

This section presents a qualitative analysis of our DR3DF-Net in comparison to 13 other excellent methods. We visualize the results restored by each algorithm and display their error maps in relation to the GT. (Lower brightness and smaller legend values in error maps correspond to closer proximity to the GT).

Figs. 4–6 present visualization results on different subsets of the StateHaze1k dataset. The DCP suffers from noticeable color distortion. The FCTF designs a two-stage dehazing model, but the high-frequency information lost in the first stage of U-Net is difficult to recover in the second stage, resulting in detail loss and haze residue. SGID and Trinity-Net introduce physical priors to enhance model interpretability, but due to the inability of parameter estimation to accurately fit complex atmospheric distributions, the dehazing effect on remote sensing images remains unsatisfactory. Restormer applies self-attention mechanisms only across multiple channels, limiting the model's expressive capability. DehazeFormer performs self-attention only within local windows, which can easily produce artifacts. PCSformer utilizes self-attention calculations only on horizontal and vertical stripes, lacking connections across stripes, which limits its modeling capabilities. FFA, based on the F-Net structure, improves recovery performance in small-scale house areas but results in darker colors in restored images. DCI-Net, PSMB-Net, and DEA-Net lose some high-frequency information during downsampling, leading to poor recovery effects in densely textured regions such as buildings and trees. TBN not only integrates an F-Net to supplement U-Net backbone details but also introduces irrelevant information, resulting in residual haze.

Compared to the methods mentioned above, our DR3DF-Net demonstrates superior dehazing performance, improving the restoration of small-scale objects such as buildings and trees while dehazing, thereby yielding satisfyingly clear dehazed images. Error maps illustrate significant differences between these algorithms and the GT, especially in areas abundant with small-scale objects like dense buildings and forests. Our results align more closely with the GT in these regions. Our DRFF can dynamically select features extracted by U-Net, F-Net, and integrate features from both networks, to effectively reconstruct clear images and restore

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.

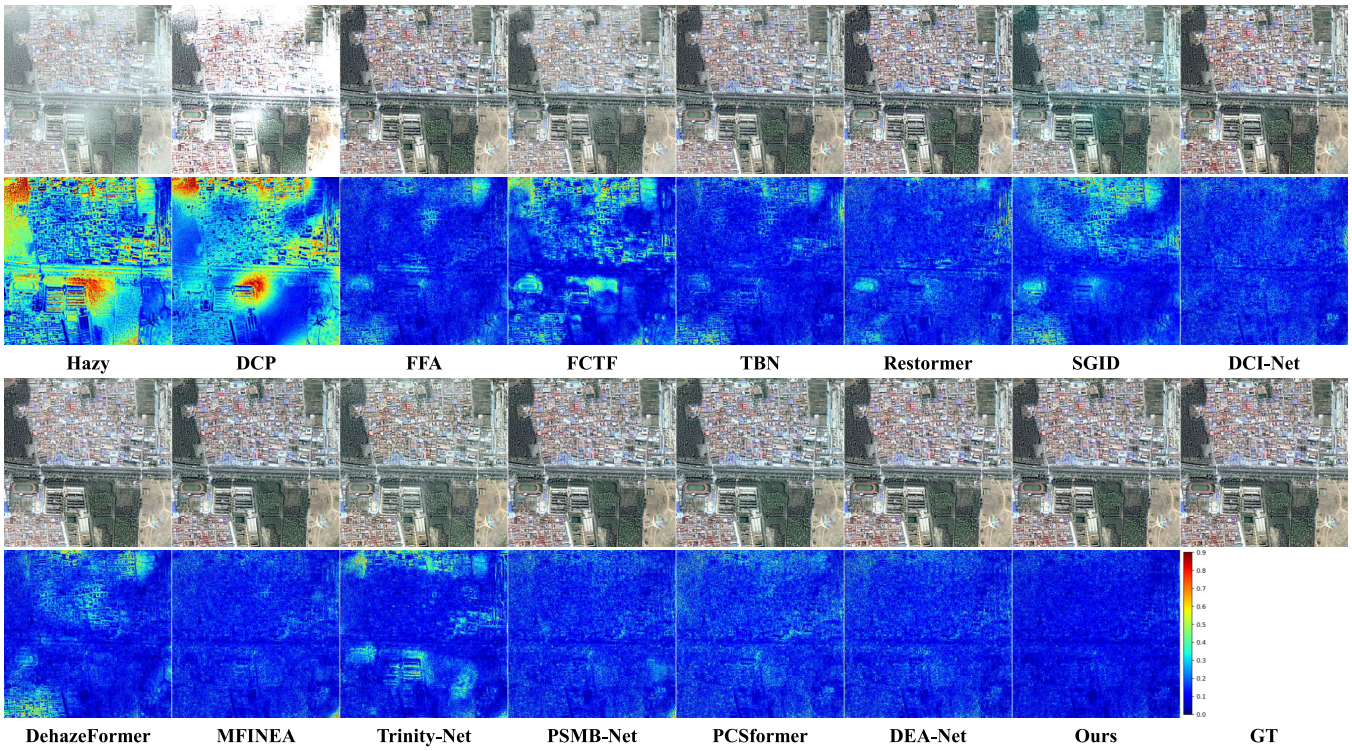


Fig. 4. Visual comparison of various haze removal methods on the SateHaze1k-Thin dataset. The first and third rows represent the visualization results of the different methods, while the second and fourth rows represent the corresponding error maps.

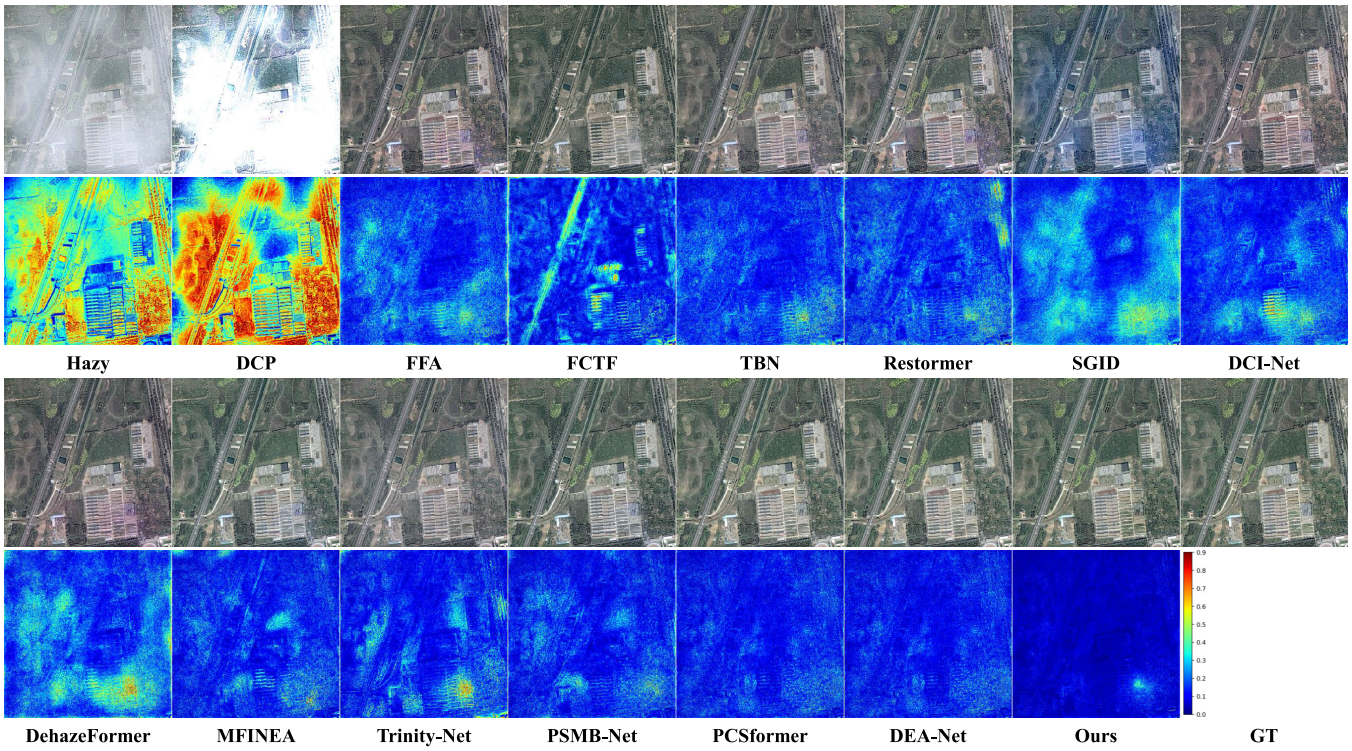


Fig. 5. Visual comparison of various haze removal methods on the SateHaze1k-Moderate dataset.

detailed information. The 3D-PFF module achieves pixel-level weight distribution through a coarse-to-fine manner during feature fusion, effectively retaining critical pixel information and enhancing the expressiveness of the fused features. As a result, it significantly improves texture details and color in dehazed images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.

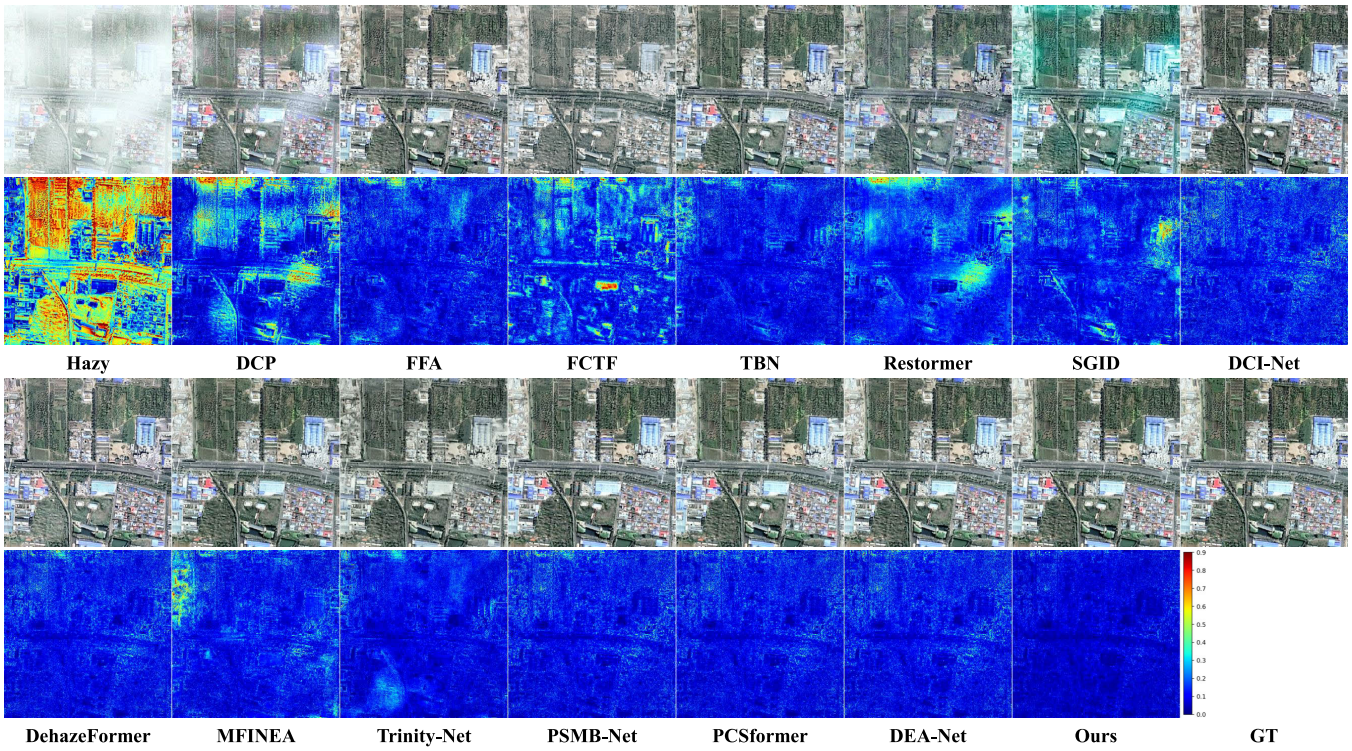


Fig. 6. Visual comparison of various haze removal methods on the SateHaze1k-Thick dataset.

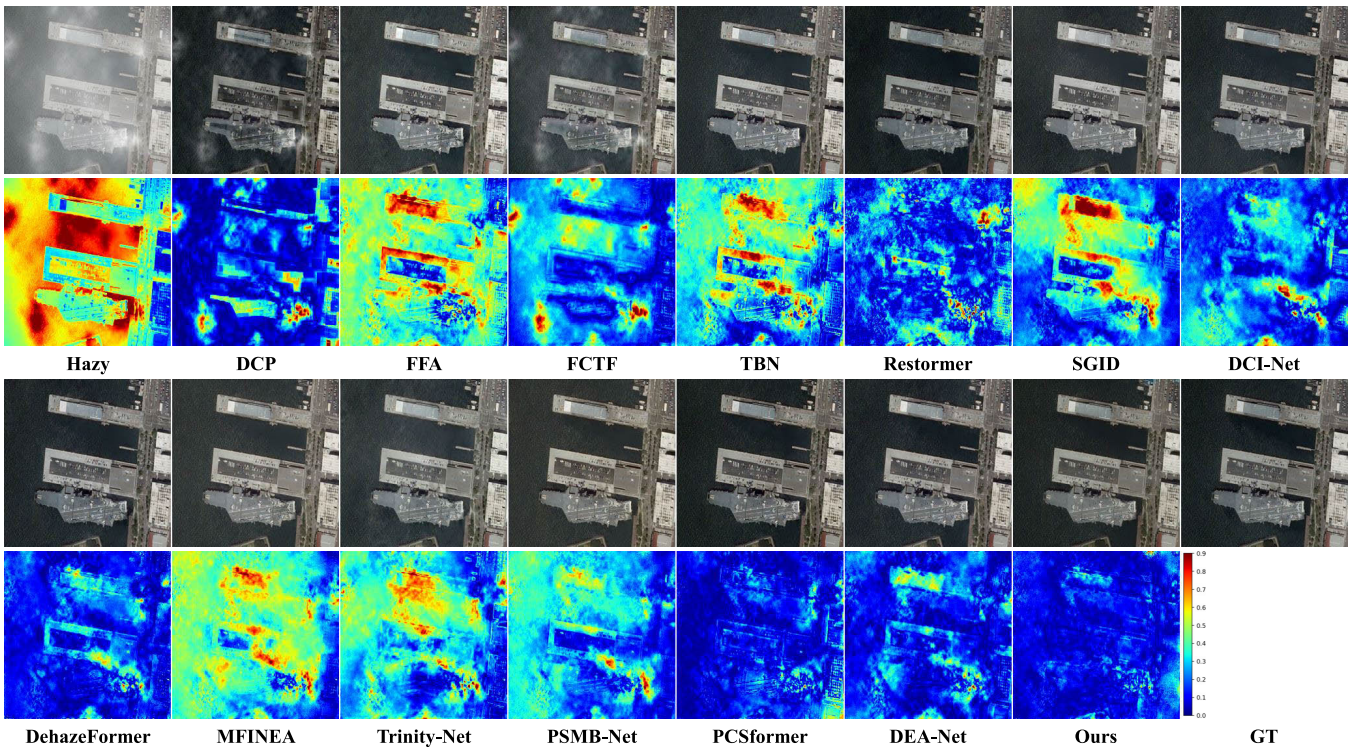


Fig. 7. Visual comparison of various haze removal methods on the RSID dataset.

Fig. 7 presents results on the RSID dataset. The haze is dense and uneven, blurring texture details. Both DCP, FCTF, and Trinity-Net leave obvious residual haze. Although Restormer, PSMB-Net, and DEA-Net perform better than DCP and FCTF, they still exhibit color deviation. By examining error maps, it is apparent that dehazed images by TBN, SGID, and DCI-Net lack detail recovery in the port and hull, whereas dehazed images by our DR3DF-Net exhibit significantly better

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.

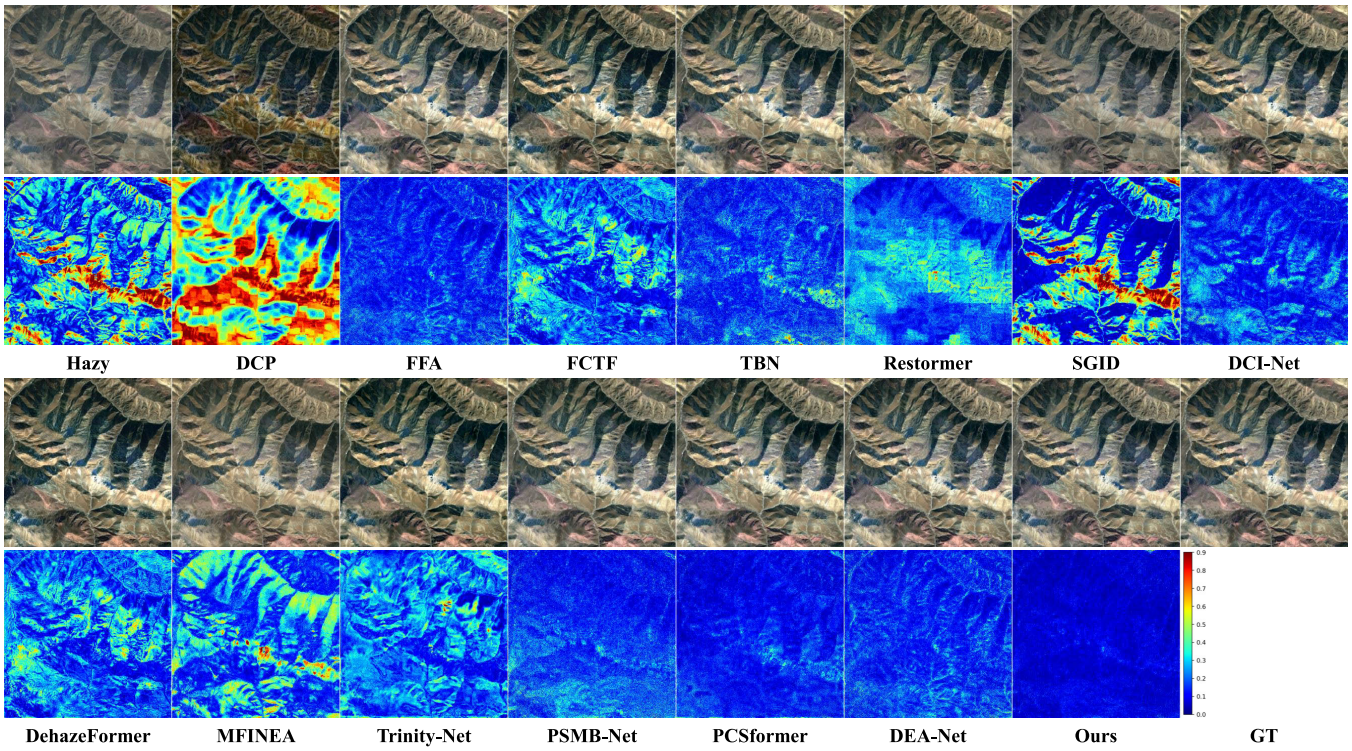


Fig. 8. Visual comparison of various haze removal methods on the RICE1 dataset.

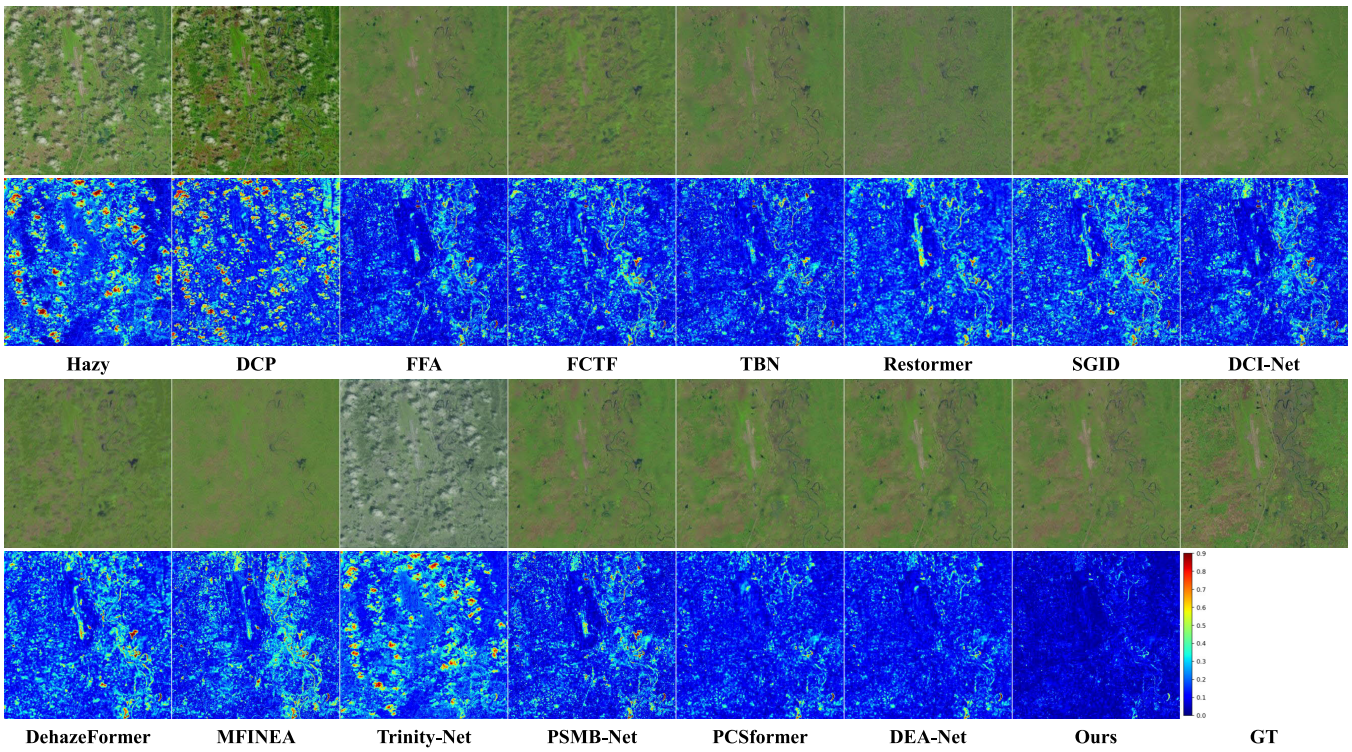


Fig. 9. Visual comparison of various haze removal methods on the RICE2 dataset.

detail recovery in these areas, and have the smallest error relative to the GT. The proposed DR3DF-Net can effectively remove haze, even in the presence of dense and uneven haze, while also successfully restoring texture details and colors.

Figs. 8 and 9 show the visualization results of all methods on the RICE dataset. On the RICE1 dataset, DCP, DehazeFormer, and Trinity-Net exhibit serious color distortion, while SGID and MFINEA suffer from significant cloud residues. In Fig. 9,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.

TABLE III

QUANTITATIVE RESULTS OF EACH INNOVATION IN THE ABLATION EXPERIMENT ON THE SateHaze1k DATASET. CONFIGS. (1)–(4) WERE CONDUCTED TO VALIDATE THE EFFECTIVENESS OF THE PROPOSED DRFF. CONFIGS. (4)–(9) WERE CONDUCTED TO VALIDATE THE EFFECTIVENESS OF THE PROPOSED 3D-PFF

Configuration	Thin		Moderate		Thick	
	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
(1) F-Net	24.21	0.909	23.86	0.901	22.23	0.872
(2) U-Net	24.12	0.904	24.57	0.903	22.68	0.870
(3) Baseline: F-Net + U-Net	25.64	0.914	25.91	0.925	23.57	0.876
(4) Baseline + DRFF	26.88	0.927	27.50	0.939	24.81	0.887
(5) Baseline + DRFF + Concat	26.61	0.924	27.30	0.942	25.17	0.889
(6) Baseline + DRFF + MixUp [35]	26.79	0.929	27.12	0.940	24.76	0.885
(7) Baseline + DRFF + SKFusion [51]	27.09	0.928	27.94	0.947	25.36	0.898
(8) Baseline + DRFF + CGAFusion [19]	27.31	0.931	28.30	0.951	25.21	0.893
(9) Baseline + DRFF + 3D-PFF	**27.70**	**0.935**	**29.04**	**0.956**	**25.72**	**0.902**

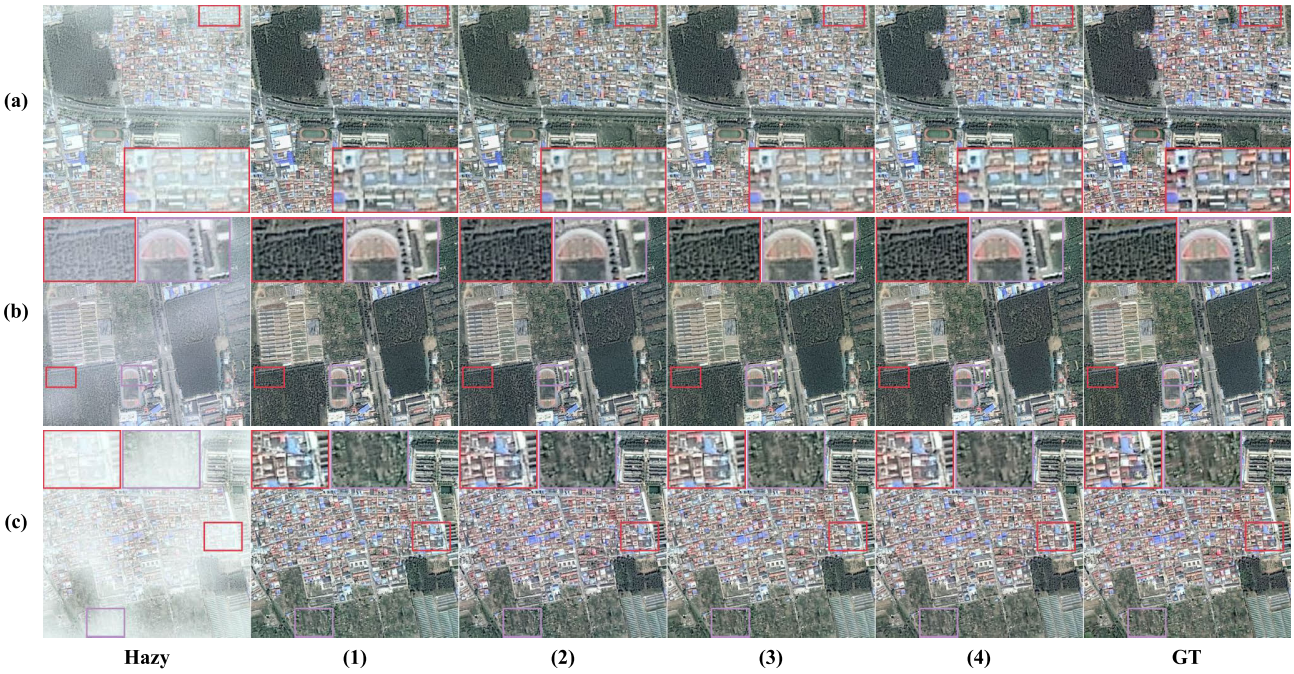


Fig. 10. Ablation results on the SateHaze1k dataset for validating the effectiveness of DRFF. (a) Thin. (b) Moderate. (c) Thick. (1) F-Net. (2) U-Net. (3) Baseline: U-Net + F-Net. (4) Baseline + DRFF.

under thick cloud conditions, DCP and Trinity-Net struggle to remove haze by estimating prior knowledge. Images reconstructed by FCTF, Restormer, SGID, DCI-Net, DehazeFormer, and MFINEA lose a significant amount of detailed information. FFA and PCSformer show some improvement in recovering details, but they still lack in color representation. Compared to PSMB-Net and DEA-net, our DR3DF-Net can effectively remove haze and demonstrate a significant improvement in texture details and color restoration. Error maps show that our method achieves notable dehazing results, producing clear images closely matching the GT in texture structure details and colors.

D. Ablation Experiments

To validate the effectiveness of DR3DF-Net, we conducted ablation experiments on the SateHaze1k dataset, focusing on the innovations of the DRFF and the 3D-PFF module. A total of nine configurations are performed: 1) F-Net, dehazing using F-Net exclusively; 2) U-Net, dehazing using U-Net exclusively; 3) baseline: F-Net + U-Net, the dehazing network is performed by integrating the two networks; 4) baseline + DRFF; 5) baseline + DRFF + Concat; 6) baseline + DRFF + MixUp; 7) baseline + DRFF + SKFusion; 8) baseline + DRFF + CGAFusion; and 9) baseline + DRFF + 3D-PFF. The specific experimental analysis is as follows.

1) Effectiveness of the DRFF: As shown in Table III, Configs. (1)–(4) were conducted to validate the effectiveness of the proposed DRFF, with their corresponding visualization results illustrated in Fig. 10. We constructed a baseline model by deploying a dual-branch network composed of U-Net and F-Net. Results from Config. (1) and Config. (2) indicated that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.

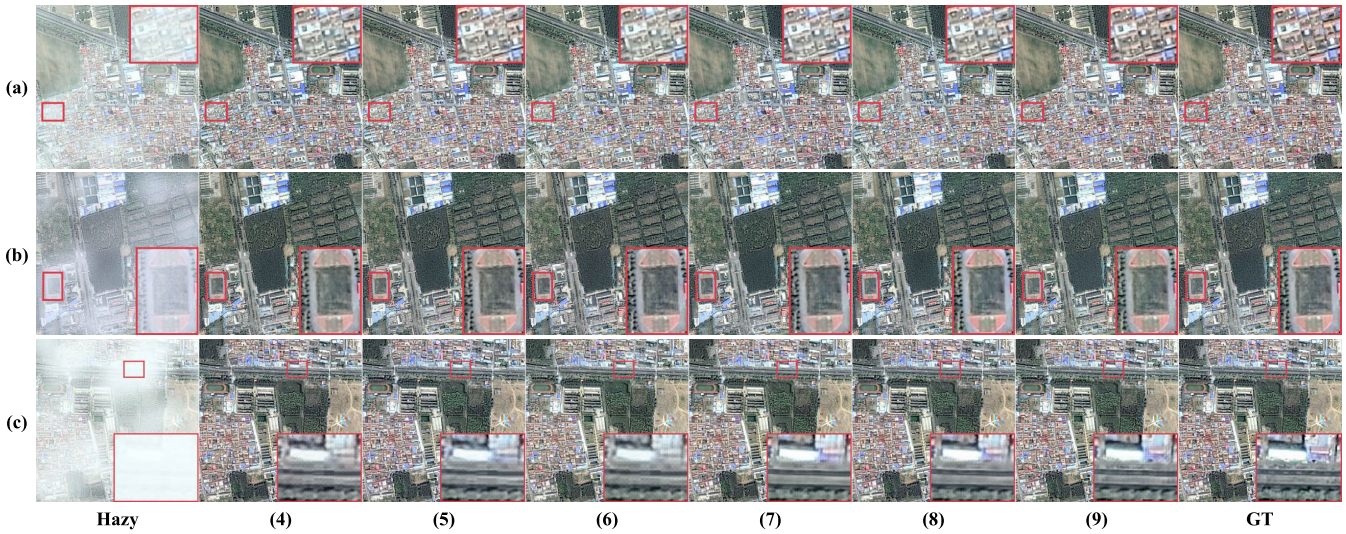


Fig. 11. Ablation results on the SateHaze1k dataset for validating the effectiveness of 3D-PFF module. (a) Thin. (b) Moderate. (c) Thick. (4) Baseline + DRFF. (5) Baseline + DRFF + Concat. (6) Baseline + DRFF + MixUp. (7) Baseline + DRFF + SKFusion. (8) Baseline + DRFF + CGAFusion. (9) Baseline + DRFF + 3D-PFF.

due to the high-frequency information loss in U-Net and the limited receptive field of F-Net, the individual haze removal performance of a single U-Net or F-Net is suboptimal. Even with the integration of two subnetworks in Config. (3), there still exist significant issues of residual haze and detail loss (as shown in Fig. 10(1)–(3)). Building upon this baseline, we developed DRFF, which contributes to notable enhancements in PSNR and SSIM metrics. The DRFF can dynamically route the extracted and fused features of U-Net and F-Net, effectively leveraging the advantages of both models in large receptive fields and detail preservation, thus enabling effective haze removal and small-scale object restoration. For instance, in the red boxes of Fig. 10(4a) and (4c), haze is effectively removed, and the restoration results on small-scale buildings are significantly improved compared to previous experiments.

2) Effectiveness of 3D-PFF Module: To validate the effectiveness of the proposed 3D-PFF module, we compared it with five fusion methods: Addition [Config. (4)], Concat, Mixup [35], SKFusion [51], and CGAFusion [19]. The quantitative results are presented in Configs. (4)–(9) of Table III, and the visual results are shown in Fig. 11. It is evident that the performance of the Addition, Concat, and MixUp fusion methods is significantly lower than that of SKFusion, CGAFusion, and our 3D-PFF module. Since SKFusion only considers the interchannel correlations, while CGAFusion primarily focuses on assigning spatial importance weight maps to each channel, neglecting the importance between channels, both result in suboptimal dehazing performance. In contrast, our approach enhances the interaction between spatial and channel information across multiple features, assigning pixel-level weights for feature fusion, which improves the texture details and color of the dehazed images, thus achieving a significant enhancement in dehazing performance. For instance, in the red box of Fig. 11(a), our method can more accurately restore the color of buildings, and in Fig. 11(c), the road edges restored by our method are clearer.

V. Discussion

Our DR3DF-Net achieves superior performance compared to several state-of-the-art methods on challenging datasets including SateHaze1k [38], RICE [39], and RSID [40]. This advantage is primarily attributed to the establishment of a feature routing framework, which enables dynamic feature selection by incorporating the SCFR. Additionally, by promoting interactions between spatial and channel dimensions to generate pixel-level weight matrices, the expressive capability of fused features is significantly enhanced. However, our method still has limitations, particularly in handling images heavily obscured by thick haze. The severe occlusion leads to substantial loss of ground information in the images, making it challenging for the model to effectively reconstruct clear images.

Recently, image inpainting [65], [66], [67] has become a popular research topic in computer vision, focusing on using deep learning models to fill in missing regions of corrupted images based on known information within the images. For instance, Liu et al. [65] utilized masks to divide missing and known regions into invalid and effective regions, thereby avoiding interference from invalid regions on feature extraction and significantly enhancing model reconstruction ability. Inspired by this field, our future work will attempt to enhance the model's ability for thick haze removal by effectively modeling the structural relationship between thick haze-obscured regions and other regions in the image, focusing on using valid information to infer the obscured regions by haze. Furthermore, we plan to extend our research to multimodal remote sensing image dehazing by incorporating multispectral information, aiming to enhance model dehazing performance and broaden its practical applications.

VI. Conclusion

In this article, we present a DR3DF-Net to address the issue of remote sensing image haze removal in challenging

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.

scenes. By integrating the DRFF and the 3D-PFF module, the performance of the proposed haze removal method can be significantly improved. It is demonstrated that the DRFF can dynamically route multiple representative features extracted via U-Net, F-Net, and their fused features by incorporating the self-generated label constrained feature routing mechanism, so as to contribute to a significant improvement in haze removal and the reconstruction of small-scale objects under haze. The 3D-PFF module can explicitly incorporate the spatial and channel dimensions to preserve significant global context and further generate pixel-level granularity weight matrices during feature fusion, enhancing feature representation and thus improving texture detail and color reconstruction. Experiments demonstrate that our DR3DF-Net outperforms several state-of-the-art methods on various benchmark remote sensing hazy datasets. Due to the challenge of effectively removing thick haze [22], our future research will focus on enhancing the model's capability for thick haze removal by efficiently modeling structural relationships between obscured and other regions. Additionally, we will explore multimodal remote sensing image haze removal, thereby enhancing the model's performance and broadening its application scope.

ACKNOWLEDGMENT

The numerical calculation of this work is supported and assisted by the Advanced Computing Center of China Three Gorges University.

REFERENCES

[1] W. Han et al., "Geological remote sensing interpretation using deep learning feature and an adaptive multisource data fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4510314.

[2] L. Zhang, L. Song, B. Du, and Y. Zhang, "Nonlocal low-rank tensor completion for visual data," *IEEE Trans. Cybern.*, vol. 51, no. 2, pp. 673–685, Feb. 2021.

[3] G. Berhan, S. Hill, T. Tadesse, and S. Atnafu, "Drought prediction system for improved climate change mitigation," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 4032–4037, Jul. 2014.

[4] A. S. Komarov, A. Caya, M. Buehner, and L. Pogson, "Assimilation of SAR ice and open water retrievals in environment and climate change Canada regional ice-ocean prediction system," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4290–4303, Jun. 2020.

[5] Z. Li et al., "In-orbit test of the polarized scanning atmospheric corrector (PSAC) onboard Chinese environmental protection and disaster monitoring satellite constellation HJ-2 A/B," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4108217.

[6] N. Jiang, H.-B. Li, C.-J. Li, H.-X. Xiao, and J.-W. Zhou, "A fusion method using terrestrial laser scanning and unmanned aerial vehicle photogrammetry for landslide deformation monitoring under complex terrain conditions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4707214.

[7] H. Chen, J. Song, C. Wu, B. Du, and N. Yokoya, "Exchange means change: An unsupervised single-temporal change detection framework based on intra- and inter-image patch exchange," *ISPRS J. Photogramm. Remote Sens.*, vol. 206, pp. 87–105, Dec. 2023.

[8] H. Chen, N. Yokoya, C. Wu, and B. Du, "Unsupervised multimodal change detection based on structural relationship graph representation learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5635318.

[9] Y. Zhang, Y. Zhao, Y. Dong, and B. Du, "Self-supervised pretraining via multimodality images with transformer for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5402711.

[10] H. Chen, N. Yokoya, and M. Chini, "Fourier domain structural relationship analysis for unsupervised multimodal change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 198, pp. 99–114, Apr. 2023.

[11] Z. Dong, M. Wang, Y. Wang, Y. Zhu, and Z. Zhang, "Object detection in high resolution remote sensing imagery based on convolutional neural networks with suitable object scale features," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 2104–2114, Mar. 2020.

[12] X. Li, J. Deng, and Y. Fang, "Few-shot object detection on remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601614.

[13] L. Chen, J. Liu, S. Sun, W. Chen, B. Du, and R. Liu, "An iterative GLRT for hyperspectral target detection based on spectral similarity and spatial connectivity characteristics," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505811.

[14] S. G. Narasimhan and S. K. Nayar, "Contrast restoration of weather degraded images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 25, no. 6, pp. 713–724, Jun. 2003.

[15] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2341–2353, Dec. 2011.

[16] D. Berman, T. Treibitz, and S. Avidan, "Single image dehazing using haze-lines," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 3, pp. 720–734, Mar. 2020.

[17] D. Berman, T. Treibitz, and S. Avidan, "Non-local image dehazing," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1674–1682.

[18] Q. Xu, Z. Wang, Y. Bai, X. Xie, and H. Jia, "FFA-Net: Feature fusion attention network for single image dehazing," in *Proc. Nat. Conf. Artif. Intell.*, vol. 34, Apr. 2020, pp. 11908–11915.

[19] Z. Chen, Z. He, and Z.-M. Lu, "DEA-net: Single image dehazing based on detail-enhanced convolution and content-guided attention," *IEEE Trans. Image Process.*, vol. 33, pp. 1002–1015, 2024.

[20] H. Sun et al., "Multi-level feature interaction and efficient non-local information enhanced channel attention for image dehazing," *Neural Netw.*, vol. 163, pp. 10–27, Jun. 2023.

[21] H. Dong et al., "Multi-scale boosted dehazing network with dense feature fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2154–2164.

[22] L. Zhang and S. Wang, "Dense haze removal based on dynamic collaborative inference learning for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5631016.

[23] B. Jiang et al., "A dehazing method for remote sensing image under nonuniform hazy weather based on deep learning network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4101717.

[24] N. Jiang, K. Hu, T. Zhang, W. Chen, Y. Xu, and T. Zhao, "Deep hybrid model for single image dehazing and detail refinement," *Pattern Recognit.*, vol. 136, Apr. 2023, Art. no. 109227.

[25] M. Xu, Y. Peng, Y. Zhang, X. Jia, and S. Jia, "AACNet: Asymmetric attention convolution network for hyperspectral image dehazing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5526714.

[26] W. Liu, X. Hou, J. Duan, and G. Qiu, "End-to-end single image fog removal using enhanced cycle consistent adversarial networks," *IEEE Trans. Image Process.*, vol. 29, pp. 7819–7833, 2020.

[27] Y. Li and X. Chen, "A coarse-to-fine two-stage attentive network for haze removal of remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 10, pp. 1751–1755, Oct. 2021.

[28] Y. Yu, H. Liu, M. Fu, J. Chen, X. Wang, and K. Wang, "A two-branch neural network for non-homogeneous dehazing via ensemble learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 193–202.

[29] Y. Liu, X. Wang, Y. Zhu, X. Fu, and Z.-J. Zha, "SDCNet: Spatially-adaptive deformable convolution networks for HR non homogeneous dehazing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 6682–6691.

[30] X. Zhang, R. Jiang, T. Wang, and W. Luo, "Single image dehazing via dual-path recurrent network," *IEEE Trans. Image Process.*, vol. 30, pp. 5211–5222, 2021.

[31] Y. Qu, Y. Chen, J. Huang, and Y. Xie, "Enhanced Pix2pix dehazing network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8152–8160.

[32] Z. Deng et al., "Deep multi-model fusion for single-image dehazing," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 2453–2462.

[33] Y. Dong, Y. Liu, H. Zhang, S. Chen, and Y. Qiao, "FD-GAN: Generative adversarial networks with fusion-discriminator for single image dehazing," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, 2020, pp. 10729–10736.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.

[34] G. Zhang, W. Fang, Y. Zheng, and R. Wang, "SDBAD-net: A spatial dual-branch attention dehazing network based on meta-former paradigm," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 1, pp. 60–70, Jan. 2024.

[35] H. Wu et al., "Contrastive learning for compact single image dehazing," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 10551–10560.

[36] H. Fu et al., "HyperDehazing: A hyperspectral image dehazing benchmark dataset and a deep learning model for haze removal," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 663–677, Dec. 2024.

[37] L. Xu, H. Li, H. Shen, C. Zhang, and L. Zhang, "A thin cloud blind correction method coupling a physical model with unsupervised deep learning for remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 246–259, Sep. 2024.

[38] B. Huang, Z. Li, C. Yang, F. Sun, and Y. Song, "Single satellite optical imagery dehazing using SAR image prior based on conditional generative adversarial networks," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2020, pp. 1806–1813.

[39] D. Lin, G. Xu, X. Wang, Y. Wang, X. Sun, and K. Fu, "A remote sensing image dataset for cloud removal," 2019, *arXiv:1901.00600*.

[40] K. Chi, Y. Yuan, and Q. Wang, "Trinity-net: Gradient-guided Swin transformer-based remote sensing image dehazing and beyond," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4702914.

[41] Q. Zhu, J. Mai, and L. Shao, "A fast single image haze removal algorithm using color attenuation prior," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3522–3533, Nov. 2015.

[42] X. Pan, F. Xie, Z. Jiang, and J. Yin, "Haze removal for a single remote sensing image based on deformed haze imaging model," *IEEE Signal Process. Lett.*, vol. 22, no. 10, pp. 1806–1810, Oct. 2015.

[43] J. Li, Q. Hu, and M. Ai, "Haze and thin cloud removal via sphere model improved dark channel prior," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 3, pp. 472–476, Mar. 2019.

[44] Y. Du, J. Li, Q. Sheng, Y. Zhu, B. Wang, and X. Ling, "Dehazing network: Asymmetric Unet based on physical model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5607412.

[45] S. Zhao, L. Zhang, Y. Shen, and Y. Zhou, "RefineDNet: A weakly supervised refinement framework for single image dehazing," *IEEE Trans. Image Process.*, vol. 30, pp. 3391–3404, 2021.

[46] H. Ullah et al., "Light-DehazeNet: A novel lightweight CNN architecture for single image dehazing," *IEEE Trans. Image Process.*, vol. 30, pp. 8968–8982, 2021.

[47] B. Li, X. Peng, Z. Wang, J. Xu, and D. Feng, "AOD-net: All-in-one dehazing network," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4780–4788.

[48] H. Zhang and V. M. Patel, "Densely connected pyramid dehazing network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3194–3203.

[49] H. Sun et al., "Partial Siamese with multiscale Bi-codec networks for remote sensing image haze removal," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4106516.

[50] Z. Lihe, J. He, Q. Yuan, X. Jin, Y. Xiao, and L. Zhang, "PhDnet: A novel physic-aware dehazing network for remote sensing images," *Inf. Fusion*, vol. 106, Feb. 2024, Art. no. 102277.

[51] Y. Song, Z. He, H. Qian, and X. Du, "Vision transformers for single image dehazing," *IEEE Trans. Image Process.*, vol. 32, pp. 1927–1941, 2023.

[52] C. Wang, J. Pan, W. Lin, J. Dong, W. Wang, and X.-M. Wu, "Self-Promer: Self-prompt dehazing transformers with depth-consistency," in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2024, vol. 38, no. 6, pp. 5327–5335.

[53] X. Zhang, F. Xie, H. Ding, S. Yan, and Z. Shi, "Proxy and cross-stripes integration transformer for remote sensing image dehazing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5640315.

[54] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 2223–2232.

[55] D. Engin, A. Genc, and H. K. Ekenel, "Cycle-dehaze: Enhanced CycleGAN for single image dehazing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 938–9388.

[56] Y. Yang, C. Wang, R. Liu, L. Zhang, X. Guo, and D. Tao, "Self-augmented unpaired image dehazing via density and depth decomposition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 2037–2046.

[57] Y. Zheng, J. Su, S. Zhang, M. Tao, and L. Wang, "Dehaze-TGGAN: Transformer-guide generative adversarial networks with spatial-spectrum attention for unpaired remote sensing dehazing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5634320.

[58] B. Li, Y. Gou, S. Gu, J. Z. Liu, J. T. Zhou, and X. Peng, "You only look yourself: Unsupervised and untrained single image dehazing neural network," *Int. J. Comput. Vis.*, vol. 129, no. 5, pp. 1754–1767, May 2021.

[59] H. Sun, Z. Luo, D. Ren, B. Du, L. Chang, and J. Wan, "Unsupervised multi-branch network with high-frequency enhancement for image dehazing," *Pattern Recognit.*, vol. 156, Dec. 2024, Art. no. 110763.

[60] J. Johnson, A. Alahi, and L. Fei-Fei, "Perceptual losses for real-time style transfer and super-resolution," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, B. Leibe, J. Matas, N. Sebe, and M. Welling, Eds., Cham, Switzerland: Springer, 2016, pp. 694–711.

[61] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[62] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, and M.-H. Yang, "Restormer: Efficient transformer for high-resolution image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5728–5739.

[63] H. Bai, J. Pan, X. Xiang, and J. Tang, "Self-guided image dehazing using progressive feature fusion," *IEEE Trans. Image Process.*, vol. 31, pp. 1217–1229, 2022.

[64] W. Zhou, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[65] G. Liu, F. A. Reda, K. J. Shih, T.-C. Wang, A. Tao, and B. Catanzaro, "Image inpainting for irregular holes using partial convolutions," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 85–100.

[66] Y. Deng, S. Hui, S. Zhou, W. Huang, and J. Wang, "Context adaptive network for image inpainting," *IEEE Trans. Image Process.*, vol. 32, pp. 6332–6345, 2023.

[67] X. Yu, L. Xu, J. Li, and X. Ji, "MagConv: Mask-guided convolution for image inpainting," *IEEE Trans. Image Process.*, vol. 32, pp. 4716–4727, 2023.



Hang Sun received the Ph.D. degree from the School of Computer Science, Wuhan University, Wuhan, China, in 2017.

He previously worked as a Senior Engineer with Huawei Technology Company Ltd., Shenzhen, China, responsible for the research and application of computer vision. He is currently an Associate Professor with the College of Computer and Information Technology, China Three Gorges University, Yichang, China. His works have been published in premier computer vision journals and conferences, including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON CYBERNETICS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, *International Journal of Remote Sensing*, *Neural Networks*, *Science China Information Sciences*, *Neural Computing and Applications*, *Chinese Journal of Electronics*, Pacific-Rim Conference on Multimedia, and International Joint Conference on Neural Networks. His research interests include image dehazing, underwater image restoration, and object detection.



Shuanglong Li is currently pursuing the M.Eng. degree with China Three Gorges University, Yichang, China.

His research interests include image processing and computer vision.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.



Bo Du (Senior Member, IEEE) received the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China, in 2010.

He is a Professor with the School of Computer Science, Institute of Artificial Intelligence, Wuhan University, where he is also the Director of National Engineering Research Center for Multimedia Software. He has more than 80 research articles published in IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON CYBERNETICS (TCYB), IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), *IEEE Journal of Selected Topics in Earth Observations and Applied Remote Sensing* (JSTARS), and IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL). Thirteen of them are Essential Science Indicators (ESI) hot articles or highly cited articles. His research interests include pattern recognition, hyperspectral image processing, and signal processing.

Dr. Du regularly serves as a Senior Program Committee (PC) Member of IJCAI and the Association for the Advancement of Artificial Intelligence (AAAI). He also serves as a reviewer of 20 Science Citation Index (SCI) magazines, including IEEE TPAMI, TCYB, TGRS, TIP, JSTARS, and GRSL. He received the Highly Cited Researcher by the Web of Science Group in 2019 and 2020, the IEEE Geoscience and Remote Sensing Society 2020 Transactions Prize Paper Award, the International Joint Conferences on Artificial Intelligence (IJCAI) Distinguished Paper Prize, the IEEE Data Fusion Contest Champion, and the IEEE Workshop on Hyperspectral Image and Signal Processing Best Paper Award in 2018. He served as an Area Chair for the International Conference on Pattern Recognition (ICPR). He serves as an Associate Editor for *Neural Networks*, *Pattern Recognition*, and *Neurocomputing*.



Lefei Zhang (Senior Member, IEEE) received the B.S. and Ph.D. degrees from Wuhan University, Wuhan, China, in 2008 and 2013, respectively.

He was a Big Data Institute Visitor with the Department of Statistical Science, University College London, London, U.K., and a Hong Kong Scholar with the Department of Computing, The Hong Kong Polytechnic University, Hong Kong, China. He is a Professor with the School of Computer Science, Wuhan University, and also with Hubei Luojia Laboratory, Wuhan. His research interests include pattern recognition, image processing, and remote sensing.

Dr. Zhang serves as a Topical Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, an Associate Editor for *Pattern Recognition*, and a Section Editor-in-Chief for *Remote Sensing*.



Dong Ren received the Ph.D. degree from the School of Agricultural Mechanization Engineering, Jilin University, Changchun, China, in 2007.

He is currently a Professor and the Deputy Director of the Department of Science and Technology, China Three Gorges University, Yichang, China. His works have been published in premier remote sensing journals and conferences, including IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and *Remote Sensing*. His research interests include artificial intelligence, remote sensing, and 3S technology.



Lyuyang Tong received the B.E. degree in computer science and technology and the B.A. degree in Japanese language from Changchun University of Technology, Changchun, China, in 2015 and 2017, respectively, the M.S. degree in software engineering from Guilin University of Technology, Guilin, China, in 2018, and the Ph.D. degree in computer science and technology from the School of Computer Science, Wuhan University, Wuhan, China, in 2022. He is currently pursuing the Ph.D. degree with the Department of Computer Science, City University of Hong Kong, Hong Kong.

He is currently a Post-Doctoral Researcher with the School of Computer Science, Wuhan University. His research interests include evolutionary computation, multiobjective optimization, and deep learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:38:10 UTC from IEEE Xplore. Restrictions apply.